To the Editors,
International Journal of Heat and Mass Transfer

Dear Editors,

Please consider our manuscript, "Field accuracy is not heat accuracy: wall-flux-constrained surrogates for steady two-dimensional non-Boussinesq convection", for publication in the *International Journal of Heat and Mass Transfer* as a full-length article.

Surrogates for thermal convection are usually trained on paired Boussinesq and compressible solutions. Public paired data sets fix the relative temperature contrast, so a network trained on them memorises one correction field and cannot rescale it at another contrast. The manuscript asks whether the metric used to select these surrogates is the right one. On 200 held-out classical-cavity samples, a model with a 0.09% median temperature-field error misses the both-wall integral Nusselt number by 2.0% on average (1.4–2.9% across seeds); in 43% of the samples on average the field error is below 1% and the Nu error above 2%. On a 22-sample random-wall replication at $\varepsilon \approx 1.2$, the corresponding field and Nu errors are 0.47% and 4.04%, with 66% on average in the same region. A wall-flux penalty in the training loss reduces the both-wall error under the training six-point stencil to 0.01% on the classical cavity and 0.58% in the replication, and the local wall-flux error to 0.08% and 1.14%; the median field error moves from 0.09% to 0.10% and from 0.47% to 0.49%. These six-point values are matched training/evaluation diagnostics against a non-converged 64^2 target, not absolute physical accuracy. The target correction carries a 0.43–0.60 percentage-point grid-refinement change, two orders of magnitude larger than 0.01%, and a stencil-free scaling model is worse than returning the Boussinesq field for Nu at four, six, eight and ten evaluation points. A complete two-seed 4×4 training/evaluation matrix puts every row minimum on its training stencil. At ten points the six-point-trained models give 1.00–2.21%, against 1.11% for returning X; the ten-point-trained models give 0.107–0.117%. The gain is therefore estimator-specific and is not presented as a universal physical improvement.

The underlying correction factorises as $D = \varepsilon P(X) + O(\varepsilon^2)$ with $\varepsilon = \Delta T/T_0$. Across 627 paired solutions, the velocity correction has nominal power-law exponents close to one, but resampling the 134 wall-mode families widens the intervals. The quadratic curvature is positive for velocity and pressure, while its magnitude and the temperature-curvature sign are unresolved. The prefactor is a functional of the flow. Training a network on $P = D/\varepsilon$ and multiplying by ε outside makes a public single-contrast data set reusable within the tested boundary-condition family, reducing the channel-averaged extrapolation error from 0.129 to 0.085. A plain model trained at the same contrast and rescaled by $\varepsilon/\varepsilon_0$ performs within 0.2% of the factorized target, so applying ε outside the learned correction, rather than the residual parameterisation, is what makes the data reusable.

The three-dimensional evidence includes both a 33-pair exponent check and a separate 15-field leave-one-contrast ROM evaluation. The 3D ROM improves the field floor by 1.26–2.40×, but it uses one resolution and does not include a wall-flux target. Alternate geometry, fluid and public-family tests are reported as consistency checks; none establishes general cross-family or three-dimensional transfer.

The study is within the journal's scope: it quantifies the departure from the Oberbeck–Boussinesq approximation, checks it on independently generated public data, and provides a surrogate design that improves wall heat transfer under a matched estimator while stating the remaining uncertainty. The main sweep is two-dimensional and steady; the 3D ROM and exponent checks are reported separately, and the remaining scope is stated explicitly. The 256^2 Boussinesq solution is within 0.37% of the Le Quéré benchmark at $Ra = 10^6$. The work is original, is not under consideration elsewhere, and all authors have approved the submission. The authors declare no competing interests. The public data are cited, and our generated data, solver configuration and reproduction scripts are available at the repository listed in the manuscript.

Yours sincerely,
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